

An Interactive Dashboard for Predicting Bank Customer Attrition

Lamees Mohammad Dalbah*, Sharaz Ali[†], Ghazi Al-Naymat[‡]

College of Engineering and IT, Ajman University, Ajman, UAE

Artificial Intelligence Research Center (AIRC), Ajman University, Ajman, UAE

Email: *Lamees.mohdalbah@gmail.com, [†]sharaz.ali@ajman.ac.ae, [‡]g.alnaymat@ajman.ac.ae

Abstract—Customer attrition rate is a significant concern for commercial banks. In this era of increased competition, banks have to compete fiercely to retain existing customers, particularly high-grade customers. Commercial banks have a clear motivation to predict customer attrition and by taking appropriate actions beforehand, they can not only significantly increase profits, but also, enhance their core competitiveness. This paper presents six machine learning algorithms, Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, Gradient Boosting Classifier and K-Nearest Neighbor, which are trained using two different publicly available datasets to predict bank customer attrition rates. The results of the algorithms are measured using four metrics: Accuracy, Precision, Recall, and F-Measure. In addition, a dashboard is designed that can provide exploratory analysis of current customers and also provides their loyalty status prediction using the mentioned machine learning algorithms. Results show that Gradient Boosting Classifier and Random Forest algorithms performed the best, reaching an average accuracy of about 87% and 97%, respectively for both datasets.

Index Terms—Machine Learning, Business Intelligence, Customer Churn, Prediction

I. INTRODUCTION

Customers today have a multitude of choices when it comes to selecting a commercial banking service and combined with the ease of access granted by information technology, it has never been easier to switch between different banking services. Therefore, commercial banks must compete fiercely to retain existing customers, particularly high-grade ones. The banking sector has concluded that its existing customer base is the most precious asset. Banks often use customer attrition rates to compare against competitors [1]. Bank Customer Attrition refers to a bank losing existing customers. A widely followed consensus among bank executives is that “it is easier to retain a customer than to acquire a new one” [2].

Primarily, customers leave their banks for four reasons – Death, Divorce, Displacement, and Dissatisfaction – Known as the 4 D’s of customer attrition and illustrated in Figure 1. Since there is not much that banks can do to stop the first three factors, they can only address the fourth factor, dissatisfaction, which accounts for the majority of attrition cases [2]. Studies in [3] have shown that by reducing customer attrition rates just by 5%, commercial banks can “increase profits by 25% to 85%”. Moreover, the cost of developing a new customer is five to six times more than retaining an old customer [4].

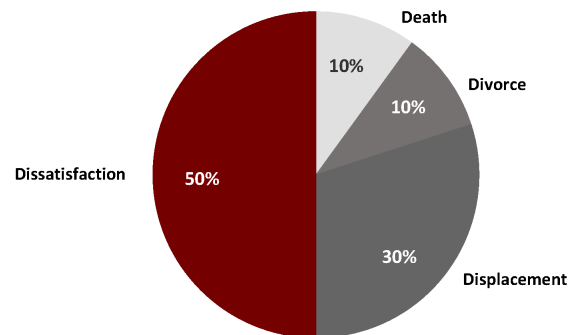


Fig. 1. The 4 D’s of The Annual Customer Attrition [2].

As a result, commercial banks have a clear motivation to predict customer attrition. By taking appropriate actions, they can significantly increase profits and enhance their core competitiveness [3]. In this context, we are about to introduce a crucial terminology in the fields of business and artificial intelligence, the term is Business Intelligence (BI). It is all about converting raw data utilizing a set of analysis methodologies in order to get insights that lead to enhance the performance of decision-making [5].

Considering the following study case, a bank manager was receiving reports which highlight a rapid increase in customer attrition, the manager was wondering about the reason behind this and how to solve it. Also, how to avoid the problem of continuous increase for the future. This paper presents six machine learning algorithms used to predict customer attrition for new bank customers and different analysis for the current customers and their loyalty status prediction. The expected objectives for such a system and desired contributions are: i) Explore the origins behind customers attrition, ii) Build an intelligent model that can predict the future churn of customers, iii) Analyze the customer behavior and establish an interactive dashboard for visualizing the analysis, iv) Increase the bank customer’s loyalty and accordingly decrease the attrition rate.

Moreover, this paper addresses some previously accomplished work in Section II, and then introduces the proposed prediction system in Section III. Consequently, the results and achievements are reported in Section IV, and finally, Section V concludes the outcomes and highlights some future work.

II. LITERATURE REVIEW

Over the past decades, researchers have used various statistical and data mining techniques to predict customer attrition. Some of the most commonly cited methods in the literature include Artificial Neural Networks, Decision Trees, Support Vector Machines, and Logistic Regression [6]. Sabbeh [7] compared ten different machine learning algorithms, including Discriminant Analysis, Decision Trees (CART), K-Nearest Neighbors, Support Vector Machines, Logistic Regression, Random Forest, ADA Boosting trees, Stochastic Gradient Boosting, Naïve Bayes, and Multi-layer perceptron, to predict customer churn on a telecommunication dataset. Results showed that random forest and ADA boost outperformed all other techniques with almost the same accuracy of 96%. Li et al. [8] utilized six different algorithms to predict bank customer attrition. These algorithms included Artificial Neural Network (ANN), Naive Bayes, K-Nearest Neighbors, Decision Tree, Logistic Regression, and RIPPER. The experimental results showed that although none of the models were perfect, the Logistic Regression, Decision Tree, and RIPPER models performed the best and were recommended for customer attrition prediction in retail banking.

Hegde and Mundada [9] proposed an Enhanced Deep Feed Forward Neural Network Model to predict customer churn. They compared their results to other algorithms such as Logistic Regression, Decision Tree, Gaussian Naïve Bayes Algorithm, and Artificial Neural Network and concluded that the Enhanced Deep Feed Forward Neural Network Model performed best in accuracy when compared with existing machine learning models. Xie et al. [10] proposed a novel learning method, Improved Balanced Random Forests (IBRF), to tackle the customer churn prediction problem. The proposed method was found to improve prediction accuracy significantly compared with other algorithms, such as Artificial Neural Networks, Decision Trees, and Class-weighted Core Support Vector Machines (CWC-SVM). Moreover, IBRF also produces better prediction results than other random forests algorithms such as balanced random forests and weighted random forests.

He et al. [3] also utilized Support Vector Machines (SVM) to predict customer attrition. In addition, the random sampling method was used to improve the performance of the learning method on imbalanced datasets. The results showed that the combination of the random sampling method and the SVM model could significantly improve the predictive power and help commercial banks predict churners more accurately. De Bock and Van den Poel [11] investigated the use of Rotation Forest and RotBoost algorithms in the customer churn prediction problem. Moreover, dimensionality reduction techniques such as Principal Component Analysis (PCA), Independent Component Analysis (ICA), and Sparse Random Projections (SRP) were compared and their effect on the algorithm performance was analyzed. Results showed that ICA-based Rotation Forests outperformed all other considered classifiers.

III. PROPOSED SYSTEM

The proposed system aims to reduce the attrition rate by predicting the current bank customers as well as the newly joining customer attrition status, Figure 2 demonstrates the system dataflow, it includes four phases which are i) Dataset loading from CSV file to the workspace. ii) Data preprocessing (blue shaded box). iii) Machine learning model training and testing (red shaded box). iv) Dashboard and model interaction (green shaded box). The upcoming sections discuss each phase extensively.

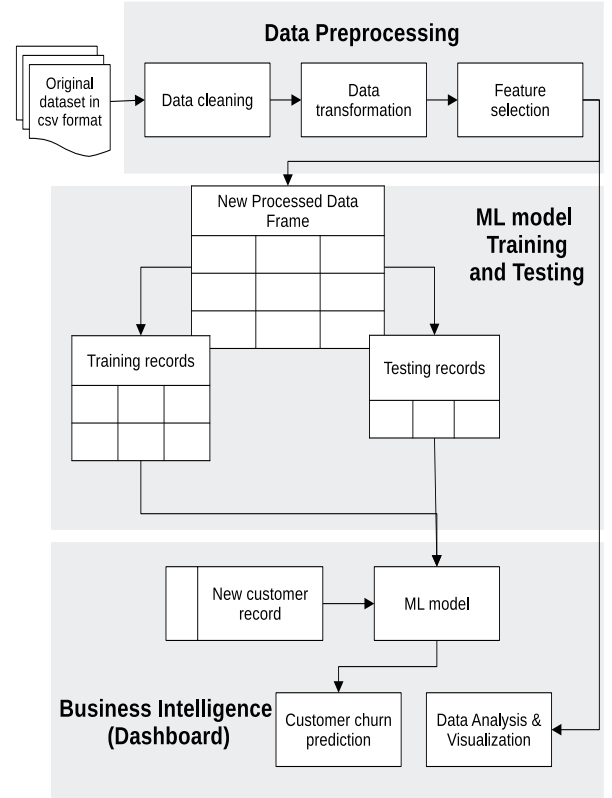


Fig. 2. The proposed system dataflow.

A. The Proposed System Dataset

Firstly and before loading the data to the workspace, the nature and features of the data that we will deal with must be understood. The proposed system was applied to two banking datasets collected from Kaggle repositories [12] [13]. The dataset features are described in Table I, in which the first dataset contains 10 thousand records. On the other hand, the second dataset has 4964 records.

B. Data preprocessing

The real-world data is not always ready to be directly imported into the machine learning algorithms. Therefore, data preprocessing takes place. The preprocessing includes removing irrelevant data, data cleaning (filling the missing data), data transformation (data encoding), and dimensionality

TABLE I
DATASET 1 & 2 FEATURES DESCRIPTION.

Dataset 1	
Feature	Description
CustomerId	Unique identifier for the customer holding the account
Surname	Customer Name
CreditScore	Number between 300-850 that depicts a consumer's creditworthiness
Geography	Customer's geographical location
Gender	Customer's gender (Male, Female)
Age	Customer's Age in Years
Tenure	Period of relationship with bank
Balance	The customer account balance
NumOfProducts	Total no. of products held by the customer
HasCrCard	0 or 1 indicates wither a customer has credit card or not (1: Yes, 0: No)
IsActiveMember	0 or 1 indicates wither a customer account is active or not (1: Yes, 0: No)
EstimatedSalary	The customer Estimated salary
Exited	The churn prediction output (0: Will not exit, 1: Will exit)
Dataset 2	
Feature	Description
CLIENTNUM	Unique identifier for the customer holding the account
Attrition_Flag	the customer attrition prediction output (Existing Customer, Attrited Customer)
Customer_Age	Customer's Age in Years
Gender	Customer's gender (M = Male, F = Female)
Dependent_coun	Number of dependents
Education_Leve	Educational Qualification of the account holder
Marital_Status	Demographic variable - Married, Single, Divorced, Unknown
Income_Category	Demographic variable - Annual Income Category of the account holder
Card_Category	Product Variable - Type of Card (Blue, Silver, Gold, Platinum)
Months_on_book	Period of relationship with bank
Total_Relationship_Count	Total no. of products held by the customer
Month_Inactive_12_mon	No. of months inactive in the last 12 months
Credit_Limit	Credit Limit on the Credit Card
Total_Revolving_Bal	Total Revolving Balance on the Credit Card
Total_Trans_Amt	Total Transaction Amount (Last 12 months)
Total_Trans_Ct	Total Transaction Count (Last 12 months)

reduction by feature selection methods. The preprocessing steps that we have implemented to the datasets are as follows

1) *Step 1 - Drop the irrelevant features:* Where the irrelevant features are the features that will not affect the prediction decision. For instance, drop the CustomerId and the Surname from dataset 1. In addition to CLENTNUM from dataset2.

2) *Step 2 - Fill the missing data:* Our Datasets include some records with missing fields, In this step two procedures have been considered to overcome this issue, the procedures are :

- Drop the records that contain missing data (e.g. Marital_Status).
- Fill the missing data with the mode of the missing feature. As an example, the mode of the Education_Level feature is Graduate. Therefore, all missing fields in Education_Level feature will be filled with Graduate. This

procedure was applied to Income_category feature of dataset 2).

3) *Step 3 - Data Encoding:* Transform the data from one data type to another by encoding the data, three encoding procedures were used to ensure type compatibility (binary encoding, ordinal encoding, and one hot encoding):

- Binary encoding: used with features such as gender, where female fields are encoded to 0 and male fields to 1. This encoding type was also used with Exited feature in dataset 1 and Attrition_flag of dataset 2.
- Ordinal encoding: used with the ordinal features such as Income_category, where each category is assigned to a single number, Less than \$40K :0, \$40K - \$60K :1, \$60K - \$80K:2, \$80K - \$120K:3, \$120K +:4. This was also applied to geography feature of dataset 1 and Education_Level feature of dataset 2.
- One hot encoding: this type of encoding creates a separated column for each unique value in the feature, then it assigns a value to 1 in case of existence, and 0 otherwise. Note that it was conducted for Marital_Status and Card_Category features from dataset 2.

4) *Step 4 - Features Scaling:* In this step, the features were scaled with a standard scalar to have all the data between - 1 and 1, for any euclidean distance calculations in the ML algorithms.

5) *Step 5 - Drop the redundant features:* One way to discover the redundant features is by calculating the correlation between the features and, accordingly the correlated features are considered as repetitive features. Consequently, eliminate the features with a correlation value of 0.8 or more for dataset1, and 0.8 or more for dataset2. Note that those values were chosen after multiple experiments. Furthermore, Figure 3 illustrates the correlations between dataset1 features, where the darker the color of the squares the higher the correlations.

C. Machine Learning Algorithms training and testing

Now the data is ready to be imported to the machine learning algorithm for training and testing phases. First, the data is splitted into training and testing. Next, the training data is utilized to train six machine learning algorithms producing six ML models. Consequently, the models will be tested using the testing data, and the best two models will be used for the system dashboard.

The six algorithms are supervised learning algorithms supplied with the labeled data. In which the algorithms are Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, Gradient Boosting Classifier, and K-Nearest Neighbor. Table II provides a brief description about each algorithm.

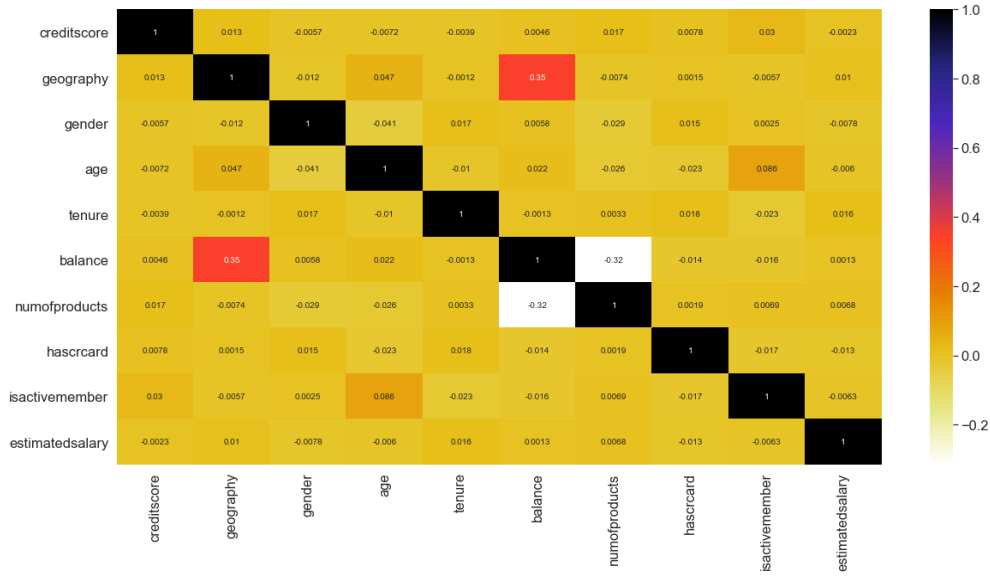


Fig. 3. The correlation measure between Dataset 1 features.

TABLE II
MACHINE LEARNING ALGORITHM DESCRIPTION.

Algorithm	Description
Logistic Regression	A predictive analysis used to explain the relationship between a dependent binary variable and a set of independent variables.
Support Vector Machine	It represents observations as points in a high-dimensional space and tries to identify the best separating hyperplanes between instances of different classes
Decision Tree	Are tree-shaped structures representing sets of decisions capable to generate classification rules for a specific dataset.
Random Forest	It extends the basic idea of a single classification tree by growing many classification trees in the training phase.
Gradient Boosting Classifier	Ensemble-based algorithms that are based on the idea of boosting, it tries to convert a set of weak learners into a stronger learner.
K-Nearest Neighbor	An algorithm that classifies a group of data points which is based on similarity measure and the value priory provided K.

D. The Business Intelligence Dashboard

To build an informative and successive communication between the data analyst and the organization manager, a user-friendly interface has been established, it displays the analysis of the bank customers. Furthermore, it predicts the loyalty level of the new joint customer and accordingly outputs a prediction about his/her attrition status in the future (using the pre-trained ML model from the previous stage).

Practically, the design of the dashboard was developed with Python using scikit-learn, dash, and plotly libraries [14]. The main steps of the dashboard creation and interaction responses are illustrated in Figure 4. Note that at any state the user can press the prediction and the data analysis tabs.

Accordingly, the required actions take place. Moreover, the flowchart reaches the end state, once the user closes the dashboard.

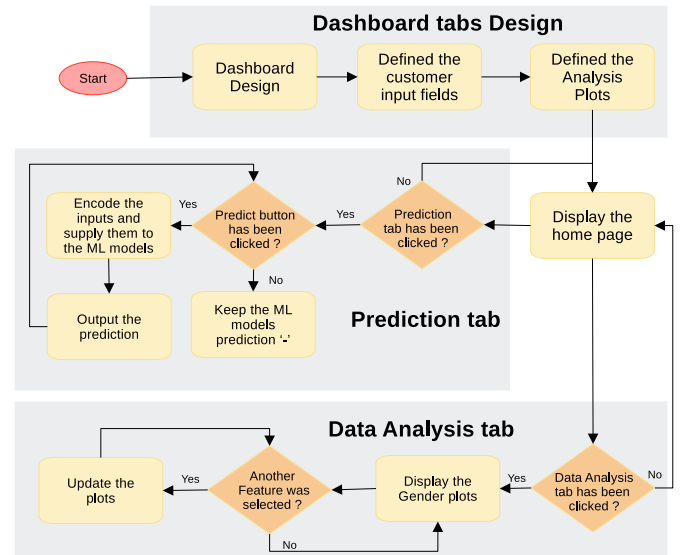


Fig. 4. The proposed system dashboard flowchart.

Where the creation process starts by declaring the HTML page design using CSS file format, as well as importing the needed figures. Next, two branches are exists according to the tab that we are considering. First, as a home page, the first appearing page is the prediction page. As long as the user enters the customer's information and presses the "Predict" button, the results of the predictions will appear, otherwise they will stay '-'. On the other hand, once clicking on the "Data Analysis" tab, the Data analysis page appears with a variety of plots controlled using the categorical fields.

More precisely, the prediction tab (Figure 5) contains several input fields that need to be filled with customer details upon clicking the 'predict' button. Consequently, the system generates a prediction of whether the customer will exit the bank or not. For the purpose of demonstration, two pre-trained models were used to provide the predictions which are the Random Forest, and Support Vector Machine (SVM). Where it is expected that both models generate the same conclusions. However, in case they differ, the decision maker will rely on the findings of the model with the higher accuracy. For instance, assume that the models SVM and Random forest had an accuracy of 80% and 90% in the learning phase, respectively. Consequently, the decision maker will consider the Random Forest result's "Exited".

Fig. 5. Dashboard design for customer churn prediction (Dataset 1).

On the other hand, the second tab contains the data exploratory analysis (Figure 6), in which several graphs are displayed (e.g. the Daunt chart, the line chart, the grouped bar chart, and the histogram). These graphs are dynamic in nature, i.e., they can be changed to display different categorical data using the drop-down menu. The data in these graphs are grouped into two categories, whether the customer exited the bank or not. Utilizing the graphs in the dashboard, decision-makers can notice patterns in customer attrition rates, whether it is a specific age group or demographic group of people that have more attrition rates. The graphs can be further customized to switch between different parameters to give a better visual representation of what factors most affect customer attrition.



Fig. 6. Dashboard design for customer data analysis (Dataset 1).

IV. RESULTS AND DISCUSSION

To test the performance of the machine learning algorithms, four performance measures were selected, Accuracy, Preci-

sion, Recall, and F-Measure. These performance measures were selected based on the most commonly cited measures in the literature for predicting customer attrition so that the results from this paper could be compared with others. In which the accuracy is defined as the ratio of correctly classified instances to the total number of instances (1). On the other hand, the precision is the ratio of instances that are truly of a class to the total instances that were classified as that class (2). Next, the recall is the percentage of total relevant results that were correctly classified by the algorithm (3). Lastly, the F-Measure is a measure that combines both the precision and the recall into an equation (4). Furthermore, the data was split with a percent of 70 to 30 as training and testing sets, respectively. In addition, it should be noted that the classifiers were modeled using Python programming language with the help of Python Scikit-learn packages [15].

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F - Measure} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

Tables III and IV show the results of the algorithm performance across both Datasets 1 and 2. Note that for determining the best algorithm we utilized the accuracy at the first stage, and then the F-Measure since it combines both the recall and the precision.

For Dataset 1 (Table III), it can be seen that the Gradient Boosting Classifier performed the best, with an accuracy score of 87% and an F-Measure score of 0.86. Followed by the random forest performed which is ranked as the second-best algorithm, with an accuracy score of 86% and F-Measure of 0.86. In contrast, among the six algorithms, logistic regression performed the worst, with an accuracy score of 81% and an F-Measure of 0.77.

On the other hand, For Dataset 2 (Table IV), the Gradient Boosting Classifier performed the best and overcomes the five algorithms, with an excellent accuracy score of 97% and an F-Measure of 0.97. Once again, the random forest also came in very close, with an accuracy of 96% and an F-Measure of 0.96. Again, Logistic regression performed the worst among the other algorithms, with an accuracy score of 90% and F-Measure of 0.9.

V. CONCLUSION

Customer attrition rates are a major concern for commercial banks. In this era of increased competition, banks have to compete fiercely to retain existing customers, particularly high-grade customers. Commercial banks have a clear motivation to predict customer attrition and by taking appropriate actions beforehand, they can not only significantly increase

TABLE III
PREDICTION RESULTS FOR DATASET 1.

Algorithm	Accuracy	Precision	Recall	F-Measure
Logistic Regression	0.81	0.78	0.81	0.77
Support Vector Machine	0.86	0.86	0.86	0.84
Decision Tree	0.81	0.81	0.81	0.81
Random Forest	0.86	0.86	0.86	0.86
Gradient Boosting Classifier	0.87	0.86	0.87	0.86
K-Nearest Neighbor	0.83	0.81	0.83	0.81

TABLE IV
PREDICTION RESULTS FOR DATASET 2.

Algorithm	Accuracy	Precision	Recall	F-Measure
Logistic Regression	0.9	0.9	0.9	0.9
Support Vector Machine	0.92	0.92	0.92	0.92
Decision Tree	0.94	0.94	0.94	0.94
Random Forest	0.96	0.96	0.96	0.96
Gradient Boosting Classifier	0.97	0.97	0.97	0.97
K-Nearest Neighbor	0.91	0.9	0.91	0.9

profits, but also enhance their core competitiveness. This paper presented six machine learning algorithms, Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, Gradient Boosting Classifier, and K-Nearest Neighbor, which were trained using two different publicly available datasets to predict bank customer attrition rates. The results of the algorithms were measured using four metrics: Accuracy, Precision, Recall, and F-Measure. Results showed that for Dataset 1, Gradient Boosting Classifier performed the best, with an accuracy score of 87% and an F-Measure score of 0.86. In addition, in the second place, we had the Random forest performed, with an accuracy score of 86% and F-Measure of 0.86. Furthermore, the same findings were achieved with the second dataset. Where the Gradient Boosting Classifier performed the best, with a leading accuracy score of 97% and an F-Measure of 0.97. Followed by the Random forest with an accuracy of 96% and F-Measure of 0.96. Using the trained models of two of the machine learning models, a business intelligence dashboard was designed that can provide customer loyalty status prediction as well as exploratory analysis of current customers for decision making. For future work, more algorithms can be included in the analysis, such as Artificial Neural Networks and Deep Learning Models, to provide readers with a much broader comparison of algorithm performance when predicting customer attrition. In addition, the dashboard can be enhanced to be data and business independent.

REFERENCES

- [1] D. Van den Poel and B. Lariviere, "Customer attrition analysis for financial services using proportional hazard models," *European journal of operational research*, vol. 157, no. 1, pp. 196–217, 2004.
- [2] A. Jennings. (2015) The 4 D's of Customer Attrition. Accessed: 2021–03–05. [Online]. Available: <https://thefinancialbrand.com/55772/banking-customer-attrition-analysis/>
- [3] B. He, Y. Shi, Q. Wan, and X. Zhao, "Prediction of customer attrition of commercial banks based on svm model," *Procedia Computer Science*, vol. 31, pp. 423–430, 2014.
- [4] L. J. Rosenberg and J. A. Czepiel, "A marketing approach for customer retention," *Journal of consumer marketing*, 1984.
- [5] N. Nithya and R. Kiruthika, "Impact of business intelligence adoption on performance of banks: a conceptual framework," *Journal of Ambient Intelligence and Humanized Computing*, vol. 12, no. 2, pp. 3139–3150, 2021.
- [6] A. Keramati, H. Ghaneei, and S. M. Mirmohammadi, "Developing a prediction model for customer churn from electronic banking services using data mining," *Financial Innovation*, vol. 2, no. 1, pp. 1–13, 2016.
- [7] S. F. Sabbeh, "Machine-learning techniques for customer retention: A comparative study," *International Journal of advanced computer Science and applications*, vol. 9, no. 2, 2018.
- [8] F. Li, J. Lei, Y. Tian, S. Punyapattanakul, and Y. J. Wang, "Model selection strategy for customer attrition risk prediction in retail banking," in *Proceedings of the Ninth Australasian Data Mining Conference-Volume 121*, 2011, pp. 119–124.
- [9] S. K. Hegde and D. Mundada, "Enhanced deep feed forward neural network model for the customer attrition analysis in banking sector," *International Journal of Intelligent Systems and Applications*, vol. 11, pp. 10–19, 07 2019.
- [10] Y. Xie, X. Li, E. Ngai, and W. Ying, "Customer churn prediction using improved balanced random forests," *Expert Systems with Applications*, vol. 36, no. 3, pp. 5445–5449, 2009.
- [11] K. W. De Bock and D. Van den Poel, "An empirical evaluation of rotation-based ensemble classifiers for customer churn prediction," *Expert Systems with Applications*, vol. 38, no. 10, pp. 12 293–12 301, 2011.
- [12] TrustTheData. (2018) Bank Customer Churn Prediction. Accessed: 2021–03–05. [Online]. Available: <https://www.kaggle.com/code/kmalit/bank-customer-churn-prediction/notebook>
- [13] T. Konstantin. (2021) Bank Churn Data Exploration And Churn Prediction. Accessed: 2021–03–05. [Online]. Available: <https://www.kaggle.com/code/thomaskonstantin/bank-churn-data-exploration-and-churn-prediction/notebook>
- [14] P. T. Inc. (2015) Collaborative data science. Montreal, QC. <https://plot.ly>. [Online]. Available: <https://plot.ly>
- [15] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.